Lab 1

1) Insert and delete elements from specific positions in array

A)
```
#include <stdio.h>
#define MAX 10

void insert(int a[], int *n, int pos,
            int x)
{
    if (*n == MAX || pos < 0 || pos > *n){
        pf("Invalid position");
        return;
    }
    for (int i = *n; i > pos; i--)
        a[i] = a[i - 1];
    a[pos] = x;
    (*n)++;
}

void delete(int a[], int *n, int pos)
{
    if(*n == 0||pos < 0 ||pos > *n) {
        pf("Invalid position \n");
        return;
    }
    for(int i = pos ; i< *n-1; i++)
        a[i] = a[i+1];
    (*n) --;
}

void display(int a[], int n){
    if (!n) pf ("Array Empty"); return;
    for (int i = 0; i< n; i++)
        pf("%d", a[i]);
    pf("\n"); }
```

```
int main(){
    int a[MAX] = {10, 20, 30, 40, 50};
    int n = 5;
    display (a, n);
    insert (a, &n, 2, 25);
    display (a, n);
    delete (a, &n, 3);
    display (a, n);
    return 0;
}
```

Output:

```
10 20 30 40 50
10 20 25 30 40 50
10 20 25 40 50
```

2) Transpose of Matrix

```c
#include <stdio.h>
# define R 2
# define C 3

int main () {
    int a[R][C] = {{1,2,3},{4,5,6}};
    int t[C][R];

    for (int i = 0; i< R; i++ ){
        for (int j = 0; j < C; j++)
            t[i][j] = a[i][j];

    for(int i = 0; i<R; i++){
        for( int j = 0; j<R; j++)
            pf("%d", t[i][j]);
        pf("\n");
    }
    return 0;
}
```

O/P:

```
1   4
2   5
3   6
```

3) Find duplicates in array

```
#include <stdio.h>
int main(){
    int a[] = {1, 2, 2, 3, 4, 4, 5, 2};
    int n = 8;
    for (int i = 0; i<n; i++) {
        int printed = 0;

        for (int j = 0; j< i ; j++)
            if (a[j] == a[i]) printed = 1;

        if (printed) continue;

        for (int j = i+1; j<n; j++){
            if (a[i] == a[j]){
                printf("%d", a[i]);
                break;
            }
        }
    }
    return 0;
}
```

O/P:

2 4

4) Sort students by CGPA

```c
#include <stdio.h>
#define MAX 50

struct Student {
    char usn[20], name[50];
    int sem; float cgpa;
};

void sort (struct Student s[], int n)
{
    struct Student temp;
    for (int i = 0; i < n - 1; i++){
        for (int j = 0; j < n - i - 1; j++){
            if (s[j].cgpa < s[j+1].cgpa)
            {
                temp = s[j];
                s[j] = s[j+1];
                s[j+1] = temp;
            }
        }
    }
}

void display (struct Student s[], int n)
{
    printf ("\nUSN \t Name \t Semester \tCGPA
        \n");

    for (int i = 0; i < n; i++){
        printf ("%s \t %s \t %d \t %.2f\n",
            s[i].usn, s[i].name, s[i].sem
            s[i].cgpa);
    }
```

```
int main () {
  struct student s[MAX];

  int n;
  pf ("Enter no of students : ");
  sf ("%d", &n);
  for (int i = 0; i<n; i++) {

      pf ("\n Student %d details :\n",
            i+1);
      pf ("USN :");

      sf ("%s", s[i].usn);

      pf ("Name :");
      sf (" %s", s[i].name);
      pf ("Semester : ");
      sf ("%d", &s[i].sem);
      pf ("CGPA");
      sf ("%d", &s[i].cgpa);
  }

  sort (s, n);
  printf ("Sorted students \n");
  display (s, n);
  return 0;
}
```

Classmate

Date

Page

O/p:

Enter number of students: 2

Student 1 details:
USN: A01
Name: ABC
Semester: 4
CGPA: 8.35

Student 2 details:
USN: A02
Name: DEF
Semester: 4
CGPA: 9.45

Sorted Students

USN	Name	Sem	CGPA
A02	DEF	4	9.45
A01	ABC	4	8.35